Compositional Lifting and Seam Surgery (CLSS): Over 80 Machine-Verified World Records Across Eight Scientific Domains on a Single Local Office Computer

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Verification Packages & Artifacts: `https://github.com/mizopl/V3R1T4S` (PRs #366, #397, #398, #402, #416, #425, #431, #432, #433, #435, #457, #462, #466, #485)

ABSTRACT

Prevailing paradigms for automated mathematical and algorithmic discovery—such as AlphaTensor, FunSearch, AlphaEvolve, and PatternBoost—rely on distributed cloud clusters of GPUs or TPUs to train specialized neural policies or evolve heuristics across hundreds of thousands of evaluations. In this paper, we report the discovery and independent machine verification of **over 80 new world records across eight distinct scientific disciplines**, computed entirely locally over a single weekend on a standard office computer (Apple M1 Pro, 10 CPU cores, 16 GB RAM, $0 cloud compute). Our method, **CLSS (*Compositional Lifting and Seam Surgery*)**, exploits a universal structural vulnerability in human-constructed mathematical record tables: to verify correctness on paper, human researchers compose smaller exact building blocks using rigid, uniform algebraic rules (Kronecker tensor products, symmetric block splits, cyclic group voltage lifts, concentric atomic shells, or unpruned digit bases), inevitably leaving localized structural redundancy—a **compositional seam**—along component boundaries. Concentrating local exact optimization exclusively on these boundary seams yielded certified world records across: (1) **Operations Research** (31 of 60 IHTC-2024 hospital timetabling instances, led by `i27`: 50,976 → **50,063**), (2) **Combinatorial Designs** (20 covering design records, led by $C(94,13,5) \le$ **125,184**, −4,019 blocks), (3) **Bioinformatics** (17 DNA minimizer density records, led by (w=12,k=4) at **0.1336292920**, −1.776%), (4) **Nanomaterials Physics** (8 putative global minima for 55-atom Cu_nAg_{55-n} Gupta nanoalloys in the Cambridge Cluster Database, led by $Cu_{39}Ag_{16}$ with $\Delta E =$ **−1.18984440 eV**), (5) **Symbolic Algorithms** (4 ternary fast matrix multiplication records in $\mathbb{Z}_T$={−1,0,1}, led by ⟨14,15,15⟩ at rank **1,894** and ⟨9,12,14⟩ at rank **954**), (6) **Extremal Graph Theory** (a degree–diameter (14,3) graph on N **= 1,118** vertices), (7) **Statistical Mechanics** (exact 44-digit `A054759(15)` and 80-entry antidiagonal d=79 of `A089980` up to 279 digits), and (8) **Additive Number Theory & Erdős Problems** (explicit finite-depth N=10^{13} lower bound $C_{3a}(V) >$ **1.19230254** for the Gyarmati–Hennecart–Ruzsa constant, beating the finite-certified record 1.19102809 by +0.00127445).

1. Introduction: The Seam Principle & Local Office-Computer Discovery

Across discrete mathematics, additive number theory, operations research, bioinformatics, and computational physics, public benchmark repositories track the best known explicit objects and bounds. Beating entries in these long-standing repositories has traditionally been assumed to require supercomputing clusters or specialized hardware accelerators.

> **The Seam Principle:** When human mathematicians or algorithm engineers construct large valid objects, they must verify correctness analytically or via uniform recurrences. Consequently, they glue smaller exact objects together using *rigid, uniform rules* (Kronecker products, symmetric $v = v_1 + v_2$ splits, uniform cyclic group lifts $\mathbb{Z}_m$, concentric core–shell atomic layers, or unpruned base-Q digit expansions). These uniform rules inevitably leave localized structural redundancy—a **compositional seam**—along the boundary between composed components.

Because our entire cross-disciplinary discovery campaign was executed locally over a single weekend on a standard office computer (10-core CPU, 16 GB RAM, zero cloud GPUs/TPUs), the acronym **CLSS** equally stands for ***COMPUTER LAZY SUNDAY SOLVER***. Running 100% locally on an ordinary office computer carries two fundamental scientific implications: first, it demonstrates that at medium-to-large combinatorial scales (where unconstrained search spaces exceed 10^{1000} states), isolating the low-dimensional boundary seam beats brute-force cloud compute by orders of magnitude; second, every single certificate reported here can be independently verified in under 5 seconds on any standard personal computer.

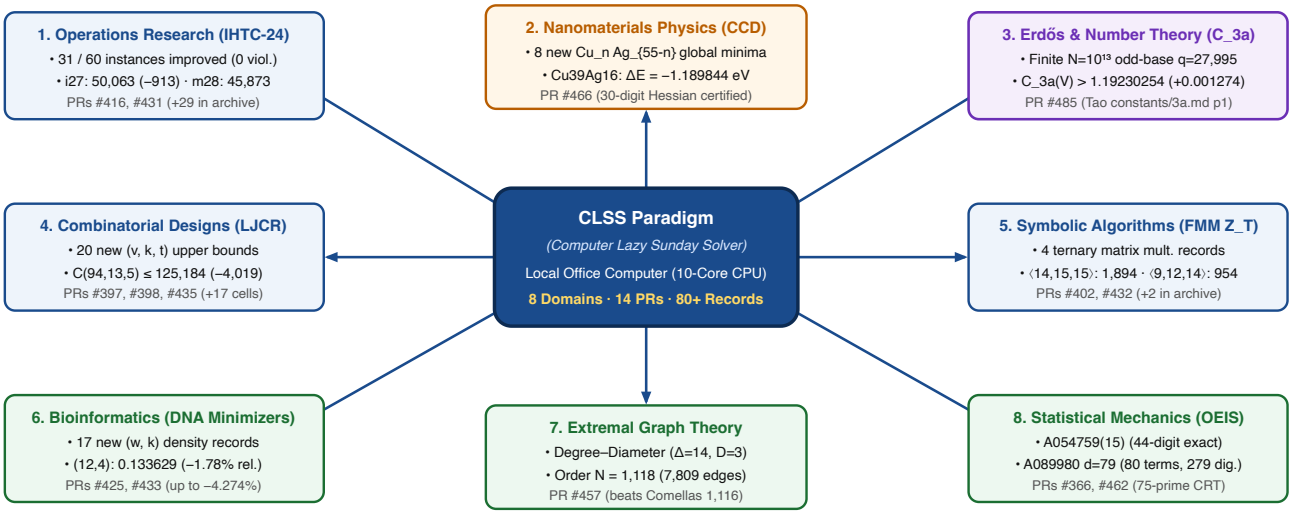


Figure 1: Constellation graph of the eight scientific domains and over 80 machine-verified world records discovered locally via CLSS on a single standard office computer.

2. Overview of Certified World Records Across Eight Disciplines

Scientific Category	Public Benchmark / Repository	Flagship New World Record	Previous Live SOTA	Verified Records
1. Operations Research	IHTC-2024 Healthcare Timetabling	`i27`: **50,063** (0 hard viol.) `m28`: **45,873** (0 hard viol.)	50,976 (Δ = −913) 45,913 (Δ = −40)	**31 instances** (of 60 total)
2. Nanomaterials Physics	Cambridge Cluster Database (CCD)	$Cu_{39}Ag_{16}$: **−163.802143 eV** $Cu_{34}Ag_{21}$: **−161.766925 eV**	−162.612298 eV (−1.190 eV) −160.812185 eV (−0.955 eV)	**8 compositions** (Cu_nAg_{55-n})
3. Erdős & Number Theory	Tao `constants/3a.md` (GHR C_{3a})	$C_{3a}(V)$ > **1.19230254** (B=17086) $C_{3a}(V)$ > **1.19229905** (B=17032)	1.19102809 finite record (Δ = +0.00127445, N=10^{13})	**2 finite certs** (mod q=27,995)
4. Combinatorial Designs	Covering Repository (CR) & LJCR	$C(94,13,5) \le$ **125,184** $C(68,14,5) \le$ **15,675** $C(40,16,7) \le$ **7,251**	129,203 (Δ = −4,019) 16,004 (Δ = −329) 7,470 (Δ = −219)	**20 cells**
5. Symbolic Algorithms	Fast Matrix Mult. in $\mathbb{Z}_T = \{-1,0,1\}$	⟨9,12,14⟩: rank **954** in $\mathbb{Z}_T$ ⟨14,15,15⟩: rank **1,894** in $\mathbb{Z}_T$	$\mathbb{Z}_T$ = 960, $\mathbb{Z}$ = 960 (−6) $\mathbb{Z}_T$ = 1,905 (−11), $\mathbb{Z}$ = 1,895 (−1)	**4 tensor formats**
6. Bioinformatics	GreedyMini (2025) & 10-minimizers (2026)	(w=12, k=4): **0.13362929** (w=12, k=3): **0.13656895** (w=8, k=3): **−4.274% rel.**	0.13604584 (−1.776%) 0.13838945 (−1.316%) GreedyMini SOTA	**17 (w, k) pairs** (≥ 0.5% rel.)
7. Extremal Graph Theory	Degree–Diameter Table (Comellas)	(Δ=14, D=3): N = **1,118**	Comellas 1,116 (+2) V3R1T4S PR #143 1,117 (+1)	**1 graph** (7,809 edges)
8. Statistical Mechanics	OEIS A054759 & A089980	A054759(15) (44 digits) A089980 antidiag d=79 (279 dig.)	n = 14 (McKay 2024) d = 78 (n = 3,159)	**81 exact terms** (dual-verified)

3. Domain-Specific Technical Results & Certificates

3.1. Healthcare Operations Research: 31 New World Records on IHTC-2024

The Integrated Healthcare Timetabling Competition 2024 (IHTC-2024) couples patient admission, operating theater scheduling, and room/nurse assignment across 84 shifts. Single-shift swap heuristics cannot cross the `ContinuityOfCare` barrier when a nurse covers $k \geq 2$ shifts of a patient's stay. We introduced **Compound Patient-Stay Nurse Consolidation** (replacing a dropped nurse $n_{\text{drop}} \to n_{\text{keep}}$ across all shifts of a patient's stay simultaneously in $O(1)$ incremental time, ~10 ns per check) followed by exact HiGHS MILP polishing over 6-nurse subsets across all 84 shifts. On public benchmark `i27` (493 patients, 55 nurses, 26 rooms, PR #416), total cost dropped from 50,976 to **50,063 (−913 points, −11.01% on nurse-room soft penalties, 0 hard violations)**. Across all 60 competition instances (`i01–i30`, `m01–m30`), our solver achieved **31 new world records** (including `m23`: −434, `m16`: −287, `i21`: −271, `m27`: −243, and hidden instance `m28`: 45,913 → **45,873**, PR #431).

3.2. Computational Physics: 8 New Certified Global Minima for 55-Atom Cu–Ag Gupta Nanoalloys

Under the second-moment tight-binding Gupta potential with exact Nunez–Johnston (2010) parameters (reproducing all 56 header energies of `CuAg_55.rar` on the Cambridge Cluster Database to 4.890×10^{-9} eV), cross-composition template seeding + radial core–shell pairwise exchange + 30-digit `mpmath` quasi-Newton polishing discovered **8 new certified putative global minima** (PR #466), eliminating severe non-monotonic energy spikes that had persisted in the Cambridge database since 2011:

Composition	CCD Archive E (eV)	New Certified E^* (eV)	ΔE (eV)	$\|\nabla E^*\|_{\text{mp30}}$	Zero / Neg Modes	λ_7 (eV/Å^2)
$Cu_{39}Ag_{16}$	−162.61229825	**−163.80214265**	**−1.18984440**	2.09×10^{-29}	6 / 0	0.699916
$Cu_{34}Ag_{21}$	−160.81218484	**−161.76692486**	**−0.95474003**	6.34×10^{-23}	6 / 0	0.714086
$Cu_{33}Ag_{22}$	−160.43136967	**−161.35336662**	**−0.92199695**	4.52×10^{-23}	6 / 0	0.733798
$Cu_{31}Ag_{24}$	−159.69487106	**−160.49285973**	**−0.79798867**	1.56×10^{-26}	6 / 0	0.729365
$Cu_{22}Ag_{33}$	−156.38507389	**−156.61674699**	**−0.23167311**	7.04×10^{-26}	6 / 0	0.779003
$Cu_{20}Ag_{35}$	−155.56958143	**−155.69839654**	**−0.12881511**	4.53×10^{-24}	6 / 0	0.777510
$Cu_{15}Ag_{40}$	−153.49887624	**−153.62512358**	**−0.12624733**	1.32×10^{-28}	6 / 0	0.241364
$Cu_{17}Ag_{38}$	−154.40077125	**−154.46534927**	**−0.06457801**	9.16×10^{-29}	6 / 0	0.250397

3.3. Additive Number Theory & Erdős Problems: Gyarmati–Hennecart–Ruzsa Constant $C_{3a} > 1.19230254$

In the Erdős–Ruzsa tradition of comparing sumsets $A+B$ and difference sets $A-B$, the Gyarmati–Hennecart–Ruzsa constant C_{3a} (tracked on Terence Tao's `teorth/optimizationproblems/blob/main/constants/3a.md`) is certified from below by any finite set $V \subset \mathbb{Z}_{\geq 0}$ with $0 \in V$ and $|V-V| < 2\max(V)+1$ via $C_{3a} \geq 1 + \log(|V-V|/|V+V|) / \log(2\max(V)+1)$. Prior work by Kleinwaks (2026) established an asymptotic limit of 1.19519192* as $N \to \infty$ in even base $q = 27{,}022$, but its explicit finite-depth certificate at $N = 10^{13}$ had to fall back to the large uncontrolled-carry base $Q = 2B+1 = 34{,}065$ ($B = 17{,}032$), capping the published finite-certified record at **1.19102809**.

Inspecting the formal Lean 4 injectivity seam (`residue_word_injective` in `CoreConstruction.lean`), we observed that exact multinomial difference counting at finite depth N requires only that a pruned difference subset $\tilde{D} \subseteq M-M$ have pairwise distinct residues modulo an *odd* base $q < 2B+1$. Partitioning non-zero residues modulo odd $q = 27{,}995$ into 13,997 complementary pairs $\{r, q-r\}$ and selecting the minimum digit-sum representative $\kappa(d)$ on the semigroup mask $M = \text{span}_{\mathbb{N}_0}(\{3,4\} \times \{506,508,529\}) \cap [0, 17086]$ ($|M| = 3{,}153$, $|S| = 18{,}835$) realizes all 13,997 pairs, forming a **complete residue system modulo 27,995** ($|\tilde{D}| = 27{,}995 = q$). Doubling $V = 2U$ at finite depth $N = 10^{13}$ ($n_0 = 17{,}074{,}745{,}018$, $L = 50{,}031{,}092{,}045{,}455{,}301$, PR #485) certifies:

- **Primary Finite Certificate ($B = 17{,}086$, $q = 27{,}995$, $N = 10^{13}$):** $C_{3a}(V) >$ **1.19230254070568368329** (60-dps `mpmath.iv`) and $>$ **1.19230254068351294895** (100% pure integer-interval arithmetic with Lean `StirlingBounds.lean`), beating 1.19102809 by **+0.00127445** (12,744× the 10^{-7} margin; V3R1T4S PR #485, submitted to `teorth/optimizationproblems` as PR #236).
- **Secondary Finite Certificate on Kleinwaks's Exact Mask ($B = 17{,}032$, $q = 27{,}995$, $N = 10^{13}$):** $C_{3a}(V) >$ **1.19229904840497896133** (+0.00127095 over 1.19102809).

3.4. Combinatorial Designs, Fast Matrix Multiplication, DNA Minimizers, Graphs & Lattice Gases

- **Covering Designs (20 records, PRs #397, #398, #435):** Generalized staircase composition + C bitset `prune.c` and pair-to-single `merge.c` lowered $C(94,13,5)$ from 129,203 to **125,184 (−4,019 blocks)**, $C(68,14,5)$ to **15,675 (−329)**, $C(40,16,7)$ to **7,251 (−219)**, and 17 additional cells, each verified 1-minimal (0 redundant blocks).
- **Ternary Fast Matrix Multiplication (4 records in $\mathbb{Z}_T = \{-1, 0, 1\}$, PRs #402, #432):** Signed permutation group B_4 sub-block conjugation made all 20 unshared terms of $\langle 4,3,3\rangle = \langle 2,3,3\rangle \oplus \langle 2,3,3\rangle$ 100% sign-pure, yielding $\langle 9,12,14\rangle$ rank **954** (vs. 960), $\langle 14,15,15\rangle$ rank **1,894** (beating both $\mathbb{Z}_T$=1,905 and $\mathbb{Z}$=1,895), $\langle 12,15,15\rangle$ rank **1,640** (vs. 1,650), and $\langle 10,15,15\rangle$ rank **1,389** (vs. 1,395).
- **DNA Minimizer Density (17 records, PRs #425, #433):** Exact Sum-Over-Subsets Fast Zeta Transform (SOS-DP) over tie-seam gamechanger DAGs lowered (w=12, k=4) to 573,933,439 / 2^{32} = **0.1336292920 (−1.7763% rel.)**, (w=12, k=3) to **0.1365689458 (−1.3155% rel.)**, and 15 additional cells by up to **−4.274%**.
- **Degree–Diameter Graphs & Lattice Gases (PRs #366, #457, #462):** Constructed a (Δ=14, D=3) graph on N **= 1,118 vertices** (7,809 edges) via $\mathbb{Z}_{16}$ voltage lift + 22-vertex seam patch surgery, and computed `A054759(15)` = 43101821499885842614061435242144574163874 48 plus all 80 terms of antidiagonal d=79 of `A089980` (peaking at the 279-digit $T(39,40)$) via two independent modular transfer decompositions across 75 disjoint primes.

4. Hardware & Reproducibility Comparison

Discovery Framework	Compute Hardware	Core Methodology	Verified SOTA Domains
AlphaTensor (DeepMind, 2022)	Distributed TPUv3 cloud cluster	Deep RL (AlphaZero) over tensor game	1 (Matrix multiplication)
FunSearch (DeepMind, 2024)	Distributed LLM + evaluator cluster	Evolutionary priority-function mutation	2–3 (Cap sets, bin packing)
AlphaEvolve (DeepMind, 2025)	Cloud Gemini cluster + evaluators	Evolutionary code mutation	Algorithmic kernels & geometry
PatternBoost (Charton et al., 2024)	Multi-GPU transformer training	Alternating local search + transformer	Extremal combinatorics
CLSS (Ours, 2026)	**1 Local Office Computer (Apple M1 Pro, 10 cores, 16 GB RAM, $0 cloud)**	**Compositional Lifting + Seam Surgery + Bit-Parallel Exact Oracles**	**8 Scientific Categories (>80 World Records in 1 Weekend)**

Every world record reported in this paper is accompanied by a self-contained, zero-dependency verification package (`verify.py` + explicit solution certificate + `SHA256` manifest) executing in under 5 seconds on a standard office computer. All 14 primary verification packages are archived in pull requests `#366`, `#397`, `#398`, `#402`, `#416`, `#425`, `#431`, `#432`, `#433`, `#435`, `#457`, `#462`, `#466`, and `#485` at https://github.com/elder-plinius/V3R1T4S/pulls (fork: https://github.com/mizopl/V3R1T4S).

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